

SMART PARKING SYSTEM WITH IOT



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SMART PARKING SYSTEM WITH IOT

Report submitted to

Department of Computer Science and Engineering

Dr. B.C. Roy Engineering College, Durgapur, WB

*for the partial fulfilment of the requirement to award the
degree of*

Bachelor of Technology

in

Computer Science and Engineering

By

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under the guidance

of

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DECLARATION

We, the undersigned, hereby declare that our B.Tech. Final Year Project entitled "**Smart Parking System with IoT**" is an original work carried out by us under the guidance of our project supervisor. To the best of our knowledge and belief, this work has not been submitted to any other institute or university for the award of any degree or diploma.

We further declare that we have not indulged in any form of plagiarism during the execution of this project or while preparing this project report. Wherever we have used materials such as data, theoretical analyses, figures, tables, or text from other sources, proper acknowledgment has been given by citing the relevant references in the report.

Finally, we undertake full responsibility for the originality and authenticity of this work and accept responsibility for its contents at any stage hereafter.

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RECOMMENDATION

This is to recommend that the work undertaken in this report entitled, “**Smart Parking System**” has been carried out by “**Anish Konar, Saptarshi Maji**” under my/our supervision and guidance during the academic year 2025-26. This may be accepted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (Computer Science and Engineering).

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APPROVAL

This is to certify that **Anish Konar, Saptarshi Maji**, students in the Department of Computer Science & Engineering, worked on the project entitled **“Smart Parking System with IoT.”**

I hereby recommend that the report prepared by them be accepted in partial fulfilment of the requirements for the Degree of Bachelor of Technology in the Department of Computer Science and Engineering, Dr. B.C. Roy Engineering College, Durgapur.

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ACKNOWLEDGEMENT

It is our privilege to express our sincere gratitude to our project supervisor, Prof. Joyjit Patra, for his valuable guidance, constant encouragement, constructive suggestions, and support throughout the course of our project.

His technical expertise, insightful feedback, and motivation have been instrumental in the successful completion of our project titled

"Smart Parking System with IoT."

We would also like to express our heartfelt gratitude to the Head of the Department, Prof. (Dr.) Raj Kumar Samanta for providing us with the opportunity, facilities, and encouragement to undertake and successfully complete this project as a part of the partial fulfilment of the requirements for the award of the B. Tech Degree.

We are sincerely thankful to all the faculty members of the Department of Computer Science and Engineering for their valuable suggestions, continuous support, and inspiration during the development of this project. Their guidance has greatly contributed to enhancing our technical knowledge and problem-solving abilities.

Finally, we extend our heartfelt thanks to our family members, friends, and everyone who directly or indirectly contributed to the successful completion of this project.

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Abstract

The increasing number of vehicles in urban areas has made parking management a major challenge, resulting in traffic congestion, fuel wastage, and environmental pollution. This project presents a Smart Parking System with IoT and Machine Learning that provides an efficient and automated parking solution. The system utilizes ESP32, ESP32-CAM, IR sensors, and servo motors to monitor parking slot occupancy, control entry and exit gates, and automate parking operations. Real-time parking information is transmitted through an IoT-based platform, enabling users to check slot availability and manage bookings using a mobile application.

To enhance security, the system employs QR code-based authentication using the ESP32-CAM for authorized vehicle access. The system also maintains parking records, including slot allocation, entry and exit times, and booking details. By reducing parking search time, improving space utilization, and providing real-time monitoring, the proposed system offers a reliable and scalable solution for modern smart city parking management.

Keywords: Smart Parking System, IoT, ESP32, QR Authentication.

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1.1 Objective

The rapid increase in the number of vehicles has made parking management a major challenge in urban areas. Traditional parking systems often require drivers to search manually for available parking spaces, leading to traffic congestion, fuel wastage, and unnecessary delays. The objective of this project is to develop an IoT-Based Smart Parking System that provides secure vehicle authentication, automated parking management, and real-time parking slot monitoring.

The proposed system consists of an ESP32 Devkit, ESP32-CAM, IR sensors, servo motors, LCD display, and LED indicators. Users can reserve a parking slot through a mobile application developed using Flutter. After a successful booking, the backend generates a unique QR code, which is displayed in the user's application. At the parking entrance, the ESP32-CAM scans the QR code and verifies it through the backend. If the QR code is valid and the entry IR sensor detects a vehicle; the entry gate opens automatically. The system continuously monitors five parking slots using IR sensors, displays parking information on the LCD screen, and automatically controls both entry and exit gates, providing a secure and efficient parking experience.

1.2 Significance

The proposed Smart Parking System offers an efficient and automated solution for modern parking management.

- **Real-Time Slot Monitoring:** Monitors the occupancy status of parking slots using IR sensors.
- **Secure Access Control:** Uses QR code authentication to ensure that only authorized vehicles enter the parking area.
- **Automated Gate Operation:** Controls entry and exit gates automatically using IR sensors and servo motors.

- **User-Friendly Mobile Application:** Allows users to book parking slots, access QR codes, and manage bookings conveniently.
- **Reduced Traffic Congestion:** Decreases the time spent searching for parking spaces, reducing congestion and fuel consumption.
- **Scalable IoT Solution:** Can be expanded to support larger parking facilities and smart city infrastructure.

1.3 Applications

The proposed IoT-Based Smart Parking System can be implemented in various environments where efficient parking management is required. Some of its major applications are:

- **Shopping malls and Commercial Complexes:** Enables customers to book parking spaces in advance, reduces waiting time at entrances, and helps manage large parking areas efficiently.
- **Hospitals and Healthcare Centres:** Provides organized parking facilities for patients, visitors, and medical staff, ensuring quicker access during emergencies and reducing congestion.
- **Educational Institutions:** Manages parking for students, faculty members, and visitors by providing secure entry through QR code authentication and real-time parking status.
- **Residential Apartments and Gated Communities:** Allows residents and authorized guests to access parking securely while preventing unauthorized vehicle entry through QR-based verification.
- **Office Buildings and Corporate Campuses:** Simplifies employee and visitor parking by enabling online slot booking, automated gate control, and efficient parking space utilization.
- **Airports, Railway Stations, and Bus Terminals:** Assists in managing high vehicle traffic by providing automated parking allocation, real-time slot monitoring, and faster vehicle movement.
- **Smart City Infrastructure:** Can be integrated with smart city transportation systems to improve parking management, reduce traffic congestion, optimize parking resources, and enhance the overall urban mobility experience.

2.1 Literature Survey

The increasing number of vehicles in urban areas has led to significant challenges in parking management. Traditional parking systems require drivers to manually search for vacant parking spaces, resulting in traffic congestion, fuel wastage, and increased travel time. To overcome these challenges, researchers have proposed various smart parking solutions using Internet of Things (IoT), wireless communication, cloud computing, and mobile applications.

Kotb et al. [1] presented an IoT-based smart parking system that uses wireless sensors to detect parking slot occupancy and provides real-time parking availability to users. Their work demonstrated that IoT technologies can significantly reduce parking search time and improve parking space utilization.

Lin et al. [2] proposed a cloud-based smart parking framework that integrates sensor networks with mobile applications. The system allows users to locate available parking spaces remotely and performs centralized parking management, improving operational efficiency.

Shinde and Patil [3] developed an RFID-based parking management system for secure vehicle entry and exit. Although the system provides effective authentication, it requires dedicated RFID cards and additional hardware, making deployment relatively expensive.

Rajabioun and Ioannou [4] introduced a reservation-based parking system that enables users to reserve parking spaces before arrival. Their approach reduced vehicle waiting time and improved parking utilization but lacked real-time monitoring of parking slot occupancy.

Recent research has also explored QR code-based authentication for parking systems. QR codes provide a secure, low-cost, and flexible method for verifying authorized vehicles without requiring specialized access cards. When integrated with IoT devices such as the ESP32-CAM, QR-based systems offer efficient vehicle authentication and automated gate control.

The proposed IoT-Based Smart Parking System builds upon these existing approaches by combining real-time parking slot monitoring using IR sensors, QR code-based vehicle authentication using ESP32-CAM, automated entry and exit gate control, and a Flutter-based mobile application for parking slot booking and QR code access.

3.1 Identification of Needs

The rapid growth of urbanization and the increasing number of vehicles have created significant challenges in parking management. Conventional parking systems often require drivers to manually search for available parking spaces, resulting in traffic congestion, increased fuel consumption, and unnecessary delays. Moreover, the absence of real-time parking information and secure access control reduces the overall efficiency of parking facilities.

To address these challenges, there is a need for an intelligent parking system that provides real-time parking slot monitoring, secure vehicle authentication, and automated gate control. The proposed IoT-Based Smart Parking System fulfils these requirements by integrating IR sensors, ESP32, ESP32-CAM, QR code authentication, and a mobile application. The system enables users to reserve parking spaces in advance, receive a unique QR code for authentication, and monitor parking availability through a user-friendly interface. This approach improves parking efficiency, enhances security, and reduces traffic congestion.

3.2 Feasibility Study

The proposed system has been evaluated based on technical, operational, and economic feasibility.

• Technical Feasibility

The system is developed using ESP32 Devkit, ESP32-CAM, Arduino IDE, Flutter, Flask, and Python. These technologies are open-source, reliable, and widely used for IoT application development. The hardware components, including IR sensors, servo motors, LCD display, and LEDs, are inexpensive and easy to integrate, making the system technically feasible.

• **Operational Feasibility**

The system provides a simple and user-friendly interface through a Flutter mobile application. Users can register, book parking slots, receive QR codes, and access parking facilities without manual intervention. Automated gate control and real-time slot monitoring improve operational efficiency while reducing human effort and parking delays.

• **Economic Feasibility**

The proposed system is cost-effective because it uses affordable hardware components and free open-source software. The implementation requires minimal maintenance and can be easily expanded to larger parking facilities, making it suitable for educational institutions, residential complexes, hospitals, and commercial buildings.

3.3 Hardware and Software Requirements

Hardware Requirements

- ESP32 Devkit
- ESP32-CAM
- IR Sensors (7)
 - 1 Entry IR Sensor
 - 5 Parking Slot IR Sensors
 - 1 Exit IR Sensor
- Servo Motors (2)
 - Entry Gate Servo
 - Exit Gate Servo
- 16×2 LCD Display with I2C Module
- Red LED & Green LED
- Breadboard and Connecting Wires
- 5V Power Supply

Software Requirements

- Arduino IDE
- Flutter SDK
- Flask Framework
- Python 3.x
- Visual Studio Code
- Android Studio (Flutter SDK support)
- PostgreSQL

4.1 System Architecture

The proposed IoT-Based Smart Parking System consists of three major components: the hardware unit, the backend server, and the mobile application. These components work together to provide real-time parking management and secure vehicle authentication.

The hardware unit includes an ESP32 Devkit, ESP32-CAM, IR sensors, servo motors, LCD display, and LED indicators. The ESP32 Devkit acts as the central controller, continuously monitoring the parking slots and controlling the entry and exit gates. The ESP32-CAM scans QR codes presented by users at the entrance for authentication.

The backend is developed using the Flask framework, which manages user accounts, parking bookings, QR code generation, and communication between the mobile application and the hardware system. The Flutter mobile application allows users to register, log in, view parking locations, book parking slots, access QR codes, and manage their profiles.

4.2 System Components

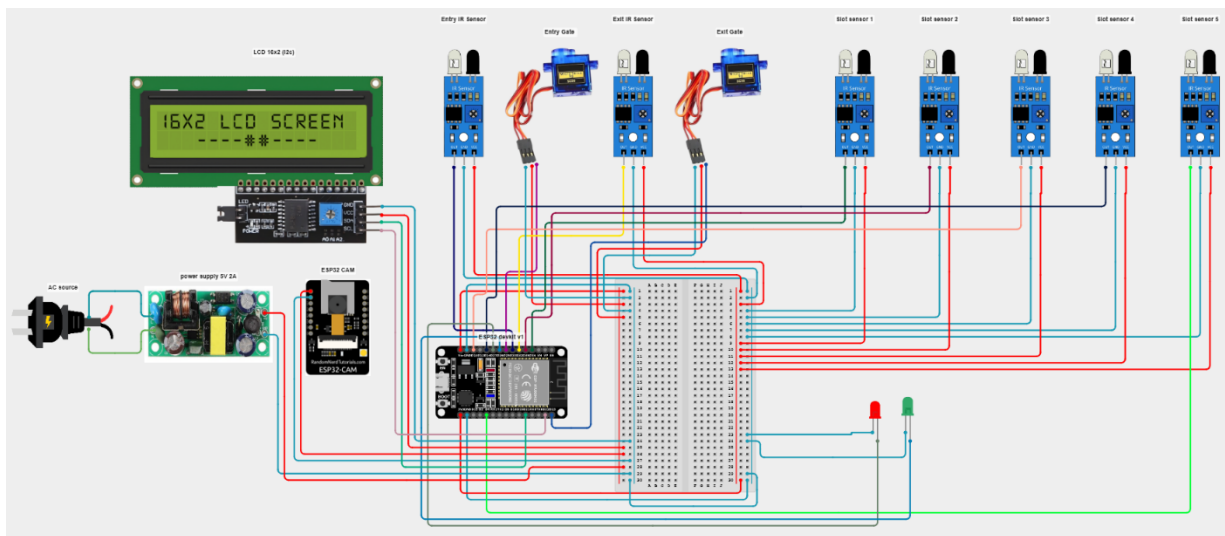
Hardware Components

- ESP32 Devkit
- ESP32-CAM
- Entry IR Sensor
- Exit IR Sensor
- 5 Parking Slot IR Sensors
- 2 Servo Motors (Entry & Exit)
- 16×2 LCD Display
- Red LED
- Green LED

Software Components

- Arduino IDE
- Python
- Flask
- Flutter
- Visual Studio Code

4.3 Hardware Circuit Diagram



The proposed hardware circuit consists of an ESP32 Devkit, which functions as the central controller of the system. It interfaces with the ESP32-CAM for QR code authentication, seven IR sensors for vehicle and parking slot detection, two servo motors for automated gate control, a 16×2 I2C LCD for displaying parking information, and red and green LEDs for system status indication. A regulated 5V, 2A power supply provides power to the entire system. The ESP32 processes data received from the sensors and communicates with the backend server to ensure secure vehicle access and real-time parking slot monitoring.

4.4 System Workflow

The proposed system operates in the following sequence:

1. The user registers and logs into the mobile application.
2. The user selects an available parking slot and confirms the booking.

3. The Flask backend stores the booking information and generates a unique QR code.
4. The generated QR code is displayed in the Active QR section of the mobile application.
5. When the vehicle reaches the parking entrance, the entry IR sensor detects its presence.
6. The ESP32-CAM scans the QR code.
7. The backend verifies the QR code.
8. If the QR code is valid, the ESP32 opens the entry gate using the servo motor.
9. The vehicle occupies the reserved parking slot.
10. The corresponding IR sensor updates the parking slot status.
11. The LCD displays the parking information, while LEDs indicate system status.
12. When the vehicle exits, the exit IR sensor detects the vehicle and the exit servo motor opens the gate automatically.

4.5 Mobile Application Design

The Flutter mobile application consists of the following modules:

- **Login Screen:** User authentication using registered credentials.
- **Dashboard:** Displays parking location details and parking availability.
- **Booking Module:** Allows users to reserve an available parking slot.
- **Active QR:** Displays the QR code generated after successful booking.
- **Profile Module:** Displays user information and account details.
- **Navigation Bar:** Provides quick navigation between Home, Booking, QR, and Profile sections.

4.6 Backend Design

The backend is developed using the Flask framework.

Its responsibilities include:

- User authentication
- Parking slot booking
- QR code generation
- QR verification
- Parking status management
- Communication with ESP32 devices

4.7 Database Design

The backend database stores all parking-related information.

Main tables include:

User Table

- User ID
- Name
- Email
- Phone Number
- Password

Parking Slot Table

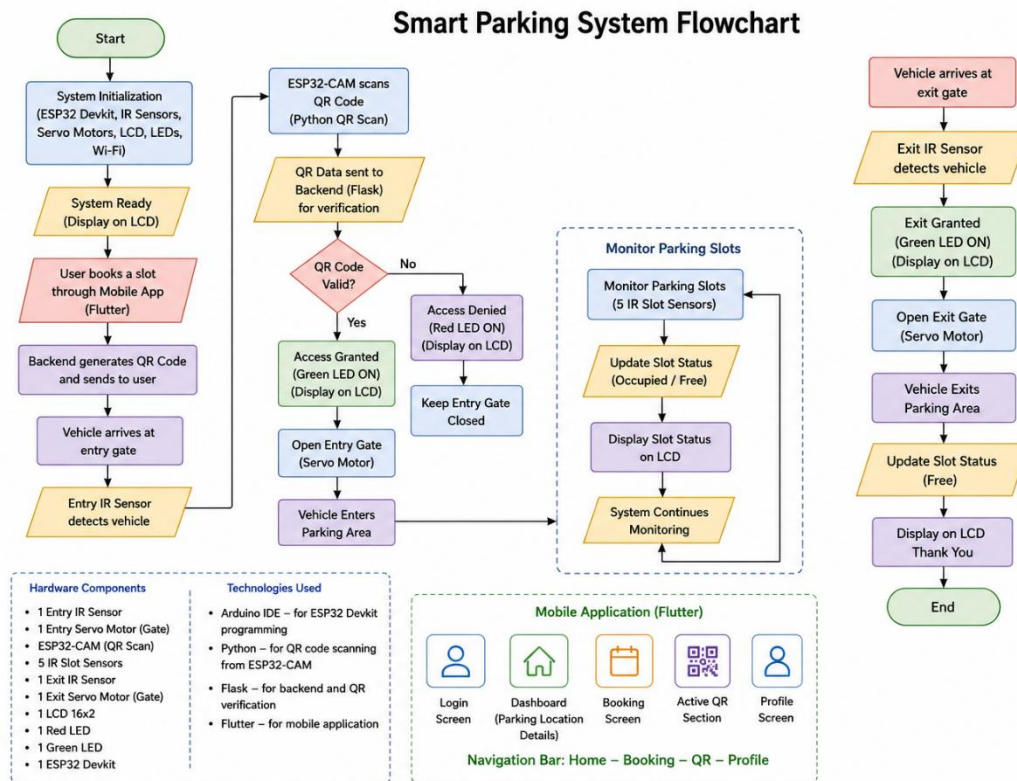
- Slot ID
- Slot Number
- Status (Available/Occupied)

Booking Table

- Booking ID
- User ID
- Slot ID
- Booking Time

- Entry Time
- Exit Time
- QR Code

4.8 Flowchart



5.1 Proposed System Workflow

The proposed IoT-Based Smart Parking System integrates IoT hardware, a backend server, and a Flutter mobile application to provide secure and efficient parking management. The complete workflow is described below.

Step 1: User Registration and Login

The user first registers through the Flutter mobile application and logs in using valid credentials. After successful authentication, the user is directed to the dashboard.

Step 2: Parking Slot Booking

The dashboard displays parking location details and the available parking slots. The user selects an available slot and confirms the booking request.

Step 3: QR Code Generation

After a successful booking, the Flask backend generates a unique QR code associated with the booking details. The generated QR code is stored in the backend database and displayed in the Active QR section of the mobile application.

Step 4: Vehicle Authentication

When the vehicle reaches the parking entrance, the entry IR sensor detects the vehicle's presence. The ESP32-CAM scans the QR code displayed on the user's mobile device. The scanned QR code is sent to the Flask backend for verification.

Step 5: Entry Gate Operation

If the QR code is valid, the backend sends an authorization response to the ESP32 Devkit. The ESP32 activates the entry servo motor, allowing the gate to open. If the QR code is invalid, the gate remains closed and access is denied.

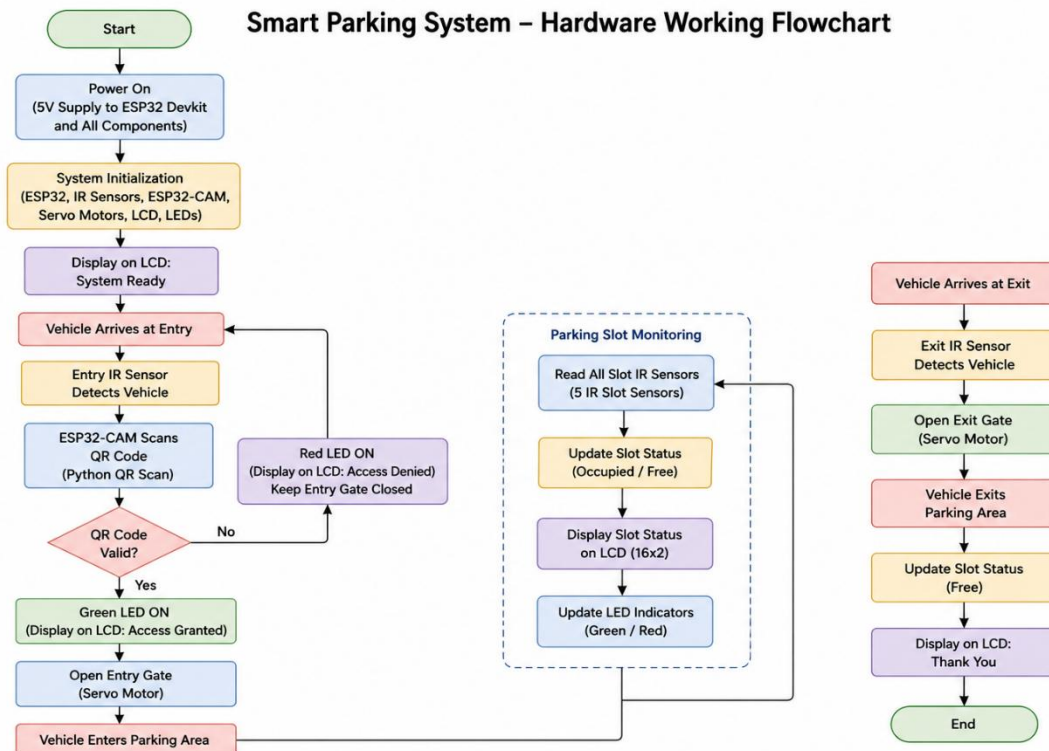
Step 6: Parking Slot Monitoring

After entering the parking area, the vehicle occupies the reserved parking slot. Five IR sensors continuously monitor the occupancy status of the parking slots. The ESP32 updates the parking status in real time and displays the information on the 16×2 LCD display.

Step 7: Vehicle Exit

When the vehicle leaves the parking area, the exit IR sensor detects the vehicle. The ESP32 automatically opens the exit gate using the exit servo motor. The parking slot is then marked as available for future bookings.

5.1.1 Proposed System Workflow Diagram



5.2 Model Evaluation and Testing

The proposed system was tested by integrating the hardware prototype with the Flask backend and Flutter mobile application. Different test cases were performed to verify the functionality of the system.

Testing Scenarios

- User registration and login.
- Parking slot booking.
- QR code generation after booking.
- QR code scanning using ESP32-CAM.
- Verification of authorized and unauthorized QR codes.

- Automatic opening of the entry gate.
- Real-time parking slot detection using IR sensors.
- Automatic opening of the exit gate.
- LCD display updates according to slot occupancy.

Performance Evaluation

The system successfully performed all major functions, including QR code authentication, automated gate control, and real-time parking slot monitoring. The communication between the mobile application, backend server, and hardware components was reliable, providing a smooth parking experience for users.

6.1 Hardware Implementation

The proposed Smart Parking System prototype was successfully implemented using an ESP32 DevKit, ESP32-CAM, IR sensors, servo motors, a 16×2 LCD display, and LED indicators. The ESP32 DevKit acts as the central controller, receiving input from the IR sensors, communicating with the backend server, and controlling the entry and exit gates. The hardware prototype operated successfully and demonstrated real-time parking management.

6.2 Mobile Application

A Flutter-based mobile application was developed to provide users with an easy-to-use interface for parking management. The application includes the following features:

• User Login	• Dashboard
• Parking Location Details	• Slot Booking
• Active QR Code	• User Profile

After booking a parking slot, the application receives a unique QR code from the Flask backend, which is used for authentication at the parking entrance.

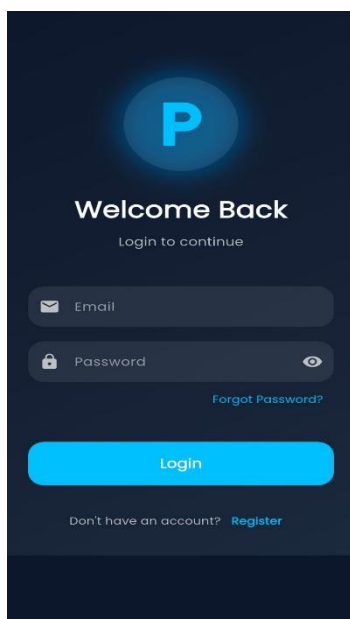


Fig. 6.1: Login Screen

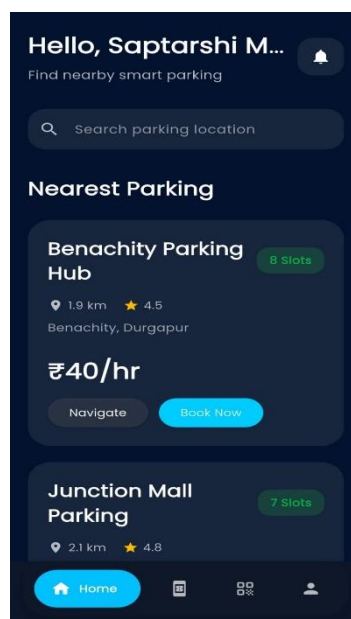


Fig. 6.2: Dashboard

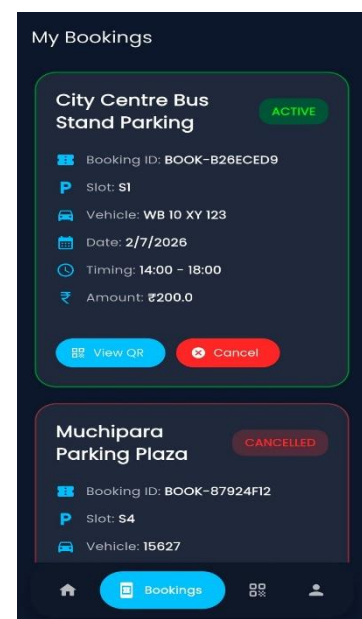


Fig. 6.3: Bookings

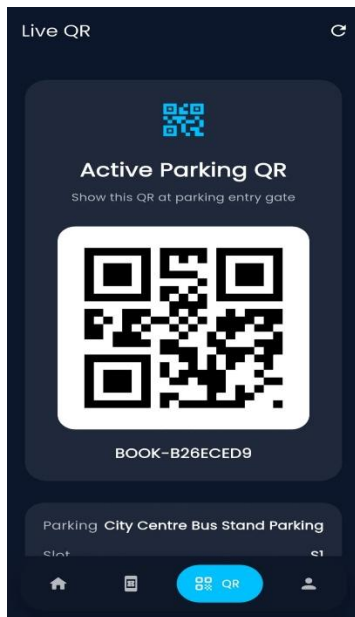


Fig. 6.4: Active QR



Fig. 6.5: Profile Screen

6.3 QR Code Authentication

The QR authentication system was tested using the ESP32-CAM. When a vehicle reaches the parking entrance, the entry IR sensor detects its presence and activates the QR scanning process. The ESP32-CAM scans the QR code displayed in the mobile application and sends the decoded information to the Flask backend for verification.

If the QR code is valid, the backend authorizes entry, and the ESP32 DevKit opens the entry gate automatically. Invalid or expired QR codes are rejected, and the gate remains closed.

Figure 6.7: QR Code Scanning Process

6.4 Parking Slot Monitoring

Five IR sensors continuously monitor the occupancy of parking slots. Whenever a vehicle occupies or vacates a parking space, the corresponding sensor updates the slot status in real time.

The LCD display provides parking information, while the LED indicators show the operational status of the system.

6.5 System Testing

The proposed system was tested under different operating conditions. The major test cases and their outcomes are summarized below.

Test Case	Expected Result	Status
User Login	Successful login	Passed
Parking Slot Booking	Slot booked successfully	Passed
QR Code Generation	QR generated successfully	Passed
QR Code Verification	Authorized QR accepted	Passed
Invalid QR Detection	Entry denied	Passed
Entry Gate Operation	Servo gate opens	Passed
Parking Slot Detection	Slot status updated	Passed
Exit Gate Operation	Exit gate opens	Passed
LCD Display	Parking status displayed	Passed

6.6 Performance Analysis

The developed system successfully performed all intended functions during testing. The QR authentication process accurately verified authorized users, while the IR sensors reliably detected vehicle occupancy in parking slots. The communication between the Flutter application, Flask backend, ESP32 DevKit, and ESP32-CAM was stable throughout the testing process.

The system demonstrated the following advantages:

- Real-time parking slot monitoring.
- Secure QR code-based vehicle authentication.
- Automatic entry and exit gate control.
- User-friendly mobile application.
- Low-cost IoT implementation.
- Easy scalability for larger parking facilities.

6.7 Result Discussion

The experimental results demonstrate that the proposed IoT-Based Smart Parking System effectively automates parking management and improves parking efficiency. Users can conveniently reserve parking spaces through the mobile application and gain secure access using QR code authentication. The integration of ESP32, ESP32-CAM, and IR sensors enables reliable real-time monitoring and gate automation. Overall, the system provides a practical, cost-effective, and scalable solution suitable for educational institutions, residential complexes, hospitals, shopping malls, and other smart parking environments.

Conclusion

The proposed **IoT-Based Smart Parking System** successfully demonstrates an efficient and automated approach to parking management using Internet of Things (IoT) technology. The system integrates an ESP32 DevKit, ESP32-CAM, IR sensors, servo motors, a 16×2 LCD display, and a Flutter-based mobile application to provide real-time parking slot monitoring and secure vehicle access. QR code-based authentication ensures that only authorized users can enter the parking area, while automated gate control enhances convenience and reduces manual intervention. The system also enables users to book parking slots in advance, view parking details, and access their QR codes through a user-friendly mobile application. Overall, the proposed solution improves parking efficiency, reduces congestion, and provides a cost-effective and scalable parking management system suitable for modern urban environments.

7.1 Key Findings

- Successfully developed an **IoT-based Smart Parking System** using ESP32 and ESP32-CAM.
- Implemented **QR code-based authentication** for secure and authorized vehicle entry.
- Developed a **Flutter mobile application** for user registration, parking slot booking, QR code access, and profile management.
- Automated entry and exit gate operation using **IR sensors** and **servo motors**.
- Achieved real-time parking slot monitoring using **five IR sensors** and displayed parking information on a **16×2 LCD**.
- Successfully integrated the hardware prototype with a **Flask backend** for booking management and QR verification.

7.2 Application Areas

The proposed system can be effectively implemented in various parking environments, including:

- Shopping malls and commercial complexes
- Hospitals and healthcare centres
- Educational institutions and university campuses
- Residential apartments and gated communities
- Office buildings and corporate campuses
- Airports, railway stations, and bus terminals
- Smart city parking management systems

7.3 Future Scope

Although the proposed system performs efficiently, several enhancements can be incorporated in future versions:

- Integrate **online payment gateways** for seamless parking fee collection.
- Implement **automatic vehicle number plate recognition (ANPR)** for additional security and faster authentication.
- Add **real-time parking notifications** to inform users about parking availability and booking status.
- Develop a **web-based admin dashboard** for monitoring parking activities, users, and booking statistics.
- Expand the system to support **multiple parking locations** with centralized management.
- Integrate **Google Maps** to guide users to their reserved parking slot and optimize navigation within large parking facilities.

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